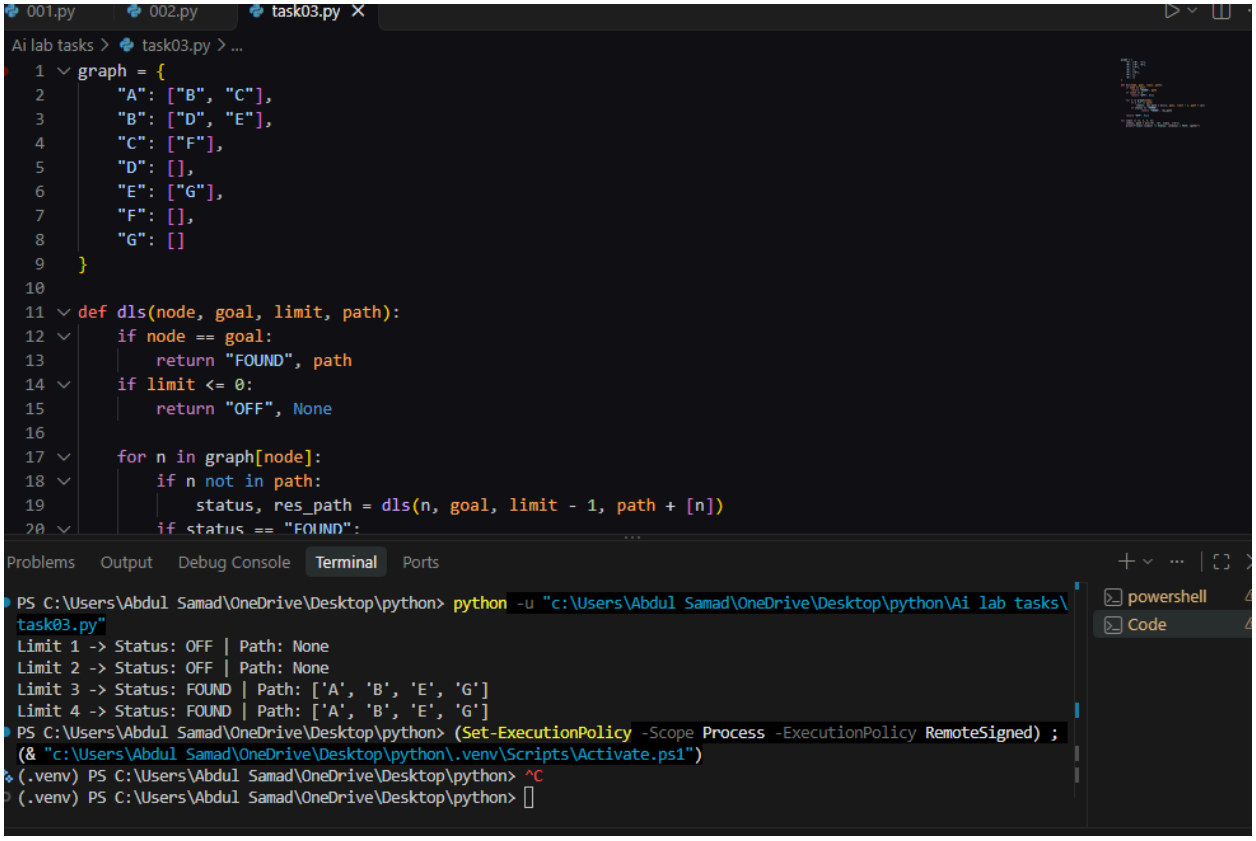


The program starts from node A and searches for goal G using Depth-Limited Search (DLS).
The `limit` decides how many levels deep the program can search.
It checks the child nodes one by one and avoids visiting nodes already in the current path.
If it finds G, it returns `FOUND` with the path; otherwise, it returns `CUTOFF`.
Limit 1: G not found. Limit 2: G not found. Limit 3: G found through `A → B → E → G`.
Limit 4: G is also found through `A → B → E → G`.